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Managed Data Services

Warehousing, pipelines, data preparation and Spark — letting Google run the infrastructure

PCA study series · 7 concept slides · 7 scenario questions

Four managed services for the data lifecycle

1 BigQuery

Serverless data warehouse: load or stream data, query petabytes with SQL, build ML models in place.

2 Dataflow

Serverless execution of Apache Beam pipelines — the same code for batch and streaming.

3 Dataprep

Visual, no-code exploration and cleaning of data; jobs run on Dataflow.

4 Managed Service for Apache Spark

Formerly Dataproc. Managed Spark and Hadoop clusters, or serverless Spark batches.

EXAM LENS Prefer the most managed option that meets the requirement — the exam rewards less operational overhead.

BigQuery essentials for architects

1 Serverless

No clusters to size. Storage and compute scale independently.

2 Pricing models

On-demand per data scanned, or capacity-based editions with reserved slots.

3 Partitioning

Split tables by date or integer range so queries scan only what they need.

4 Clustering

Sort data within partitions by columns you filter on, cutting scans further.

5 Streaming

Ingest rows continuously for near-real-time dashboards.

6 BigQuery ML

Train and predict with SQL, without moving data out.

EXAM LENS 'Reduce query cost' → partition and cluster, and select only the columns you need.

How a Dataflow pipeline moves data

1

Read

From Pub/Sub, Cloud Storage, databases or Kafka.

2

Transform

Parse, filter, enrich, join and aggregate with Beam transforms.

3

Window

Group unbounded streams into time windows; handle late data with triggers.

4

Write

To BigQuery, Bigtable, Cloud Storage or Pub/Sub.

5

Operate

Workers autoscale; templates let others launch pipelines without code.

EXAM LENS 'Real-time stream processing, autoscaling, no clusters' → Pub/Sub + Dataflow + BigQuery.

Dataprep puts data cleaning in analysts' hands

Analysts explore and fix data visually; the service turns their steps into scalable jobs.

Profile	See distributions, missing values and mismatched types at a glance.
Transform	Build a recipe of suggested or custom steps without writing code.
Run	Recipes execute as Dataflow jobs over full datasets.
Deliver	Write clean data to BigQuery or Cloud Storage on a schedule.

Running Spark and Hadoop the cloud way

1 Fast, short-lived clusters

Clusters start in about 90 seconds, so create one per job and delete it afterwards instead of keeping it idle.

2 Separate storage from compute

Keep data in Cloud Storage rather than HDFS, so it survives cluster deletion.

3 Cut cost

Add Spot VMs as secondary workers for fault-tolerant processing.

4 Serverless Spark

Submit batches without managing clusters at all.

EXAM LENS 'Migrate existing Spark or Hadoop jobs with minimal changes' → Managed Service for Apache Spark.

Dataflow or Managed Spark?

Dataflow

Choose for new, serverless pipelines

- Unified batch and streaming in one model
- No clusters; workers autoscale
- Strong streaming semantics: windows, triggers, exactly-once
- Best for new development on Google Cloud

Managed Service for Apache Spark

Choose to reuse the Hadoop ecosystem

- Runs existing Spark, Hive, Pig or Hadoop code
- Familiar open-source tools and libraries
- Clusters or serverless batches
- Best for lift-and-shift of current jobs

EXAM LENS The tie-breaker is usually existing code: have Spark jobs → Spark service; starting fresh → Dataflow.

Decision cheat sheet: data services

If the scenario says…	Reach for
Ad-hoc SQL over terabytes, no infrastructure	BigQuery
Queries keep scanning whole tables	Partition and cluster in BigQuery
Real-time stream processing with autoscaling	Dataflow (often fed by Pub/Sub)
Analysts clean data visually, no code	Dataprep
Existing Spark or Hadoop jobs	Managed Service for Apache Spark
Batch Spark jobs at lowest cost	Ephemeral clusters, Cloud Storage, Spot secondary workers
Simple ML on data already in the warehouse	BigQuery ML



7 PCA-style scenarios

Read every requirement word, eliminate options that break one, then pick the most managed answer that satisfies them all. Each question is followed by its answer and the reasoning.

Question 1 of 7

Hadoop migration

A company runs hundreds of Spark and Hive jobs on an on-premises Hadoop cluster. It wants to move to Google Cloud quickly with as few code changes as possible.

A Rewrite the jobs as Apache Beam pipelines on Dataflow

B Run the jobs on Managed Service for Apache Spark, with data moved to Cloud Storage

C Load all data into Cloud SQL and convert jobs to SQL

D Recreate the Hadoop cluster on self-managed Compute Engine VMs

Answer 1: B

B Run the jobs on Managed Service for Apache Spark, with data moved to Cloud Storage

Why it fits

The managed Spark service runs the same Spark and Hive code, and storing data in Cloud Storage decouples it from cluster lifetime.

Why the others don't

A — A full rewrite, the opposite of minimal change.

C — Wrong tool for big data processing.

D — Possible, but keeps all the cluster operations burden.

EXAM LENS 'Existing Spark/Hadoop' + 'minimal changes' → Managed Spark.

Question 2 of 7

Streaming

Clickstream events arrive through Pub/Sub at variable rates. You need 1-minute aggregates written to BigQuery with automatic scaling and no clusters to manage.

A. Dataflow streaming pipeline with fixed 1-minute windows

B. A Spark cluster polling Pub/Sub every minute

C. Cloud Run function writing each event to Cloud SQL

D. Dataprep scheduled every hour

Answer 2: A

A Dataflow streaming pipeline with fixed 1-minute windows

Why it fits

Dataflow is serverless, autoscales with load and has native windowing for streaming aggregation into BigQuery.

Why the others don't

B — Cluster management and polling add operations work and latency.

C — Per-event writes to a relational DB won't scale and don't aggregate.

D — Batch, hourly and aimed at analyst wrangling.

EXAM LENS 'Streaming' + 'windows' + 'serverless' → Dataflow.

Question 3 of 7

Data preparation

Marketing analysts receive messy CSV exports each week. They need to fix formats, remove duplicates and load results into BigQuery — and they don't write code.

A Dataprep

B Managed Service for Apache Spark

C Cloud Shell scripts

D Bigtable

Answer 3: A

A Dataprep

Why it fits

Dataprep provides a visual interface for profiling and cleaning data, then runs the recipe at scale and writes to BigQuery.

Why the others don't

B — Requires writing Spark code.

C — Requires scripting skills.

D — A NoSQL database, not a preparation tool.

EXAM LENS 'Non-technical users' + 'clean data visually' → Dataprep.

Question 4 of 7

Warehouse

A retailer wants analysts to run ad-hoc SQL over 80 TB of sales history, with no servers to size, patch or scale.

A. Cloud SQL for PostgreSQL with read replicas

B. BigQuery

C. Bigtable

D. Spanner

Answer 4: B

B BigQuery

Why it fits

BigQuery is a serverless warehouse built for interactive SQL analytics over very large datasets.

Why the others don't

A — Not designed for analytical scans at this size.

C — No SQL analytics engine; key-based access.

D — Built for transactional workloads, not warehousing.

EXAM LENS 'Analytics' + 'SQL' + 'TBs' + 'serverless' → BigQuery.

Question 5 of 7

Query cost

Most BigQuery queries filter on order_date for the last 7 days and on store_id, but each query scans the full 40 TB table. Costs are climbing.

A. Move the table to Cloud Storage and query it as external data

B. Partition the table by order_date and cluster by store_id

C. Add more slots to the reservation

D. Export the data to Cloud SQL nightly

Answer 5: B

B Partition the table by order_date and cluster by store_id

Why it fits

Partitioning lets queries prune to the dates they need, and clustering narrows scans within each partition — far less data processed.

Why the others don't

A — External tables usually scan more and run slower.

C — Faster but not cheaper per scan.

D — Unsuited to this volume and query pattern.

EXAM LENS 'Filters on date' + 'scans whole table' → partition (+ cluster).

Question 6 of 7

Spark cost

A nightly Spark job runs for 2 hours. The cluster stays up all day, and HDFS on the cluster holds the input data. You need to cut costs significantly.

A Keep the cluster and switch to larger machines

B Store data in Cloud Storage, create a cluster per job, use Spot secondary workers and delete it when the job ends

C Convert the job to BigQuery scheduled queries without changes

D Move the cluster to a region with cheaper storage

Answer 6: B

B Store data in Cloud Storage, create a cluster per job, use Spot secondary workers and delete it when the job ends

Why it fits

Ephemeral clusters bill only during the job, Cloud Storage keeps data independent of the cluster, and Spot workers lower compute price.

Why the others don't

A — Still pays for 22 idle hours.

C — Spark code can't run as scheduled queries without rewriting.

D — Doesn't address idle compute.

EXAM LENS 'Idle cluster' → ephemeral clusters + Cloud Storage + Spot.

Question 7 of 7

ML in place

A data team with strong SQL skills wants a churn-prediction model on customer data that already lives in BigQuery, without exporting data or managing ML infrastructure.

A Export to CSV and train on a laptop

B BigQuery ML

C Bigtable

D Managed Spark cluster with custom MLlib code

Answer 7: B

B BigQuery ML

Why it fits

BigQuery ML trains and serves models with SQL inside the warehouse, avoiding data movement and infrastructure.

Why the others don't

A — Insecure, unscalable and manual.

C — Storage, not a modelling tool.

D — Works, but adds code and cluster management the team doesn't need.

EXAM LENS 'SQL skills' + 'data in BigQuery' + 'ML' → BigQuery ML.